Wuao Liu

Contact

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Research Interests

My research focuses on computer vision and machine learning, with an emphasis on learning visual representations from multimodal supervision—integrating images, audio, and text. I am particularly interested in self-supervised and contrastive learning methods for visual recognition, detection, and segmentation. My work spans applications in biology and healthcare, with a growing interest in medical image analysis, especially in segmentation and representation learning for microscopy and histopathology data.

Education

University of Massachusetts Amherst — Aug. 2024 – Apr. 2028 (expected)
Ph.D. in Computer Science, GPA: 3.85 / 4.00 — Amherst, MA

University of Michigan — Aug. 2021 – Apr. 2023
M.S. in Robotics, GPA: 4.00 / 4.00 — Ann Arbor, MI

Zhejiang University — Sep. 2017 – Jun. 2021
B.Eng. in Robotics, GPA: 3.89 / 4.00 — Hangzhou, China

Publications

1. Mustafa Chasmai, **Wuao Liu**, Subhransu Maji, and Grant Van Horn. **Audio Geolocation: An Investigation with Natural Sounds**. *Under Review*.
2. Filippos Bellos, Yayuan Li, **Wuao Liu**, and Jason Corso. **Can Large Language Models Reason About Goal-Oriented Tasks?** In *Proceedings of ACL Workshop on the Scaling Behavior of Large Language Models*, pages 24-34, 2024. [paper]

Work Experience

University of Michigan — Jun. 2023 – Apr. 2024
Research Engineer — Ann Arbor, MI
Mentor: Dr. Jason Corso

- Developed a perception system for AR glasses to assist users with cooking tasks. [project]
- Designed and optimized a data collection protocol, resulting in a curated dataset of **224** instructional videos.
- Implemented a Faster R-CNN object detector to identify **14** kitchen items, achieving a mean Average Precision (mAP) of **85.99** on the curated dataset.
- Implemented a transformer-based model for procedural step recognition, achieving **92%** accuracy and enabling automated analysis of step transitions.

Research Experience

University of Massachusetts Amherst — Jan. 2025 – Apr. 2025
Advisor: Dr. Grant Van Horn — Amherst, MA
Project: **Weakly Supervised Learning for Natural World Data Analysis**

- Extended the Masked Autoencoder (MAE) framework to the audio domain, enabling self-supervised representation learning through spectrogram reconstruction.
- Achieved a **1.6%** improvement in accuracy on a downstream sound identification task.

University of Massachusetts Amherst — Aug. 2024 – Dec. 2024
Advisor: Dr. Grant Van Horn — Amherst, MA
Project: **Contrastive Learning of Audio and Geographical Data**

- Proposed the first work to geolocate natural audio at a global scale.
- Implemented a retrieval-based approach integrating audio geolocation and species range prediction, achieving a localization error of **1127** km and a regional-level localization accuracy of **17.2%**.

Technical Skills

- Programming Languages: Python, C / C++, MATLAB, LaTeX.
- Deep Learning Frameworks: PyTorch, TensorFlow, Hugging Face.
- Machine Learning Packages: Scikit-learn, JSON, Pandas